Chunked XYZZ batch inversion: report

Question and scope

How many outputs should share one batch inversion in the fixed-radix-14 XYZZ evaluator? The benchmark-only overlay splits the frozen 128-input list into chunks and invokes the checked parent batch-inversion implementation on each chunk. It accepts only sizes that divide 128, so each input appears exactly once per pass and output order remains the input order. The underlying scalar path and point table are unchanged. The prefix/reverse inversion identity is proved in the parent experiment; the native holdout and full-output replay check the specific implementation.

Direct sizing screen

The exploratory screen tested 1, 2, 4, 8, 16, 32, 64 and 128 outputs per inversion over 96 passes on the same fixture; all 12,288 results per invocation were verified. Online times were respectively 141.963, 97.671, 78.819, 68.423, 64.508, 62.930, 61.925 and 61.658 ms. This is an uncalibrated selector screen, not the qualified comparison.

Frozen paired panels

The frozen panels each use 16 alternating repetitions of A/A, comparison, A/A on the same 128-scalar fixture, with adjacent A/A acceptance range `[0.90, 1.11]`. All 96 arms and 48 pairs per panel passed; all outputs matched, artifacts stayed stable, and no A/A failure occurred.

Outputs per inversion	Online mean ms (reference / candidate)	Paired median ratio [95% interval]
1	135.419257 / 136.136782	0.992657× [0.991476, 0.994458]
2	135.681652 / 98.151946	**1.379409× [1.375878, 1.387388]**

The one-output result is a qualified 0.74% throughput regression; the two-output result is a qualified 37.94% throughput gain.

The one-output panel does not improve throughput. Two outputs per inversion is the smallest tested batch size with a qualified gain for this frozen fixture and Tier 1 instance. The size-2 candidate uses 64 inversions per 128-output pass instead of 128. Both candidates retain 26,035,988 bytes; table preparation is reported separately. Neither result establishes single-call cold latency or end-to-end application cost. Physical-host exclusivity is unverified.

Audit and implementation review

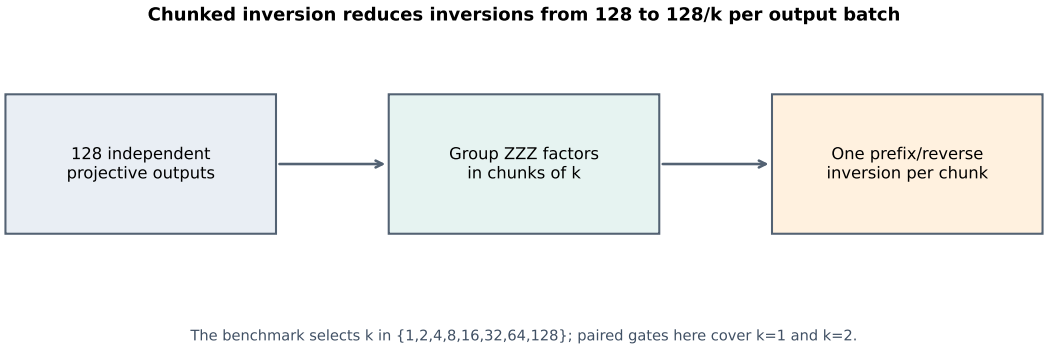
`analyze_crossover_tier1.py` regenerated `tier1-crossover-analysis.json` byte-for-byte. It checks both frozen manifests, all 192 arm outputs, both 48-pair ledgers, per-arm binary hashes, the candidate source and exact binary provenance. Source review confirmed that the overlay rejects sizes outside `[1,128]` and non-divisors of 128, applies `.chunks(batch_size)` once in each pass, and concatenates each chunk's results in order. It retains the original 128-output CLI by dispatching it to size 128. No production path is altered by the benchmark-only overlay.

The prior experiment's canonical scalar-pipeline costs are unchanged because this measures only XYZZ recovery batch size; it is not an end-to-end scalar multiplication result. The report graph is local to this stage comparison and does not change the pipeline dashboard. The 1-output regression remains visible alongside the 2-output improvement.

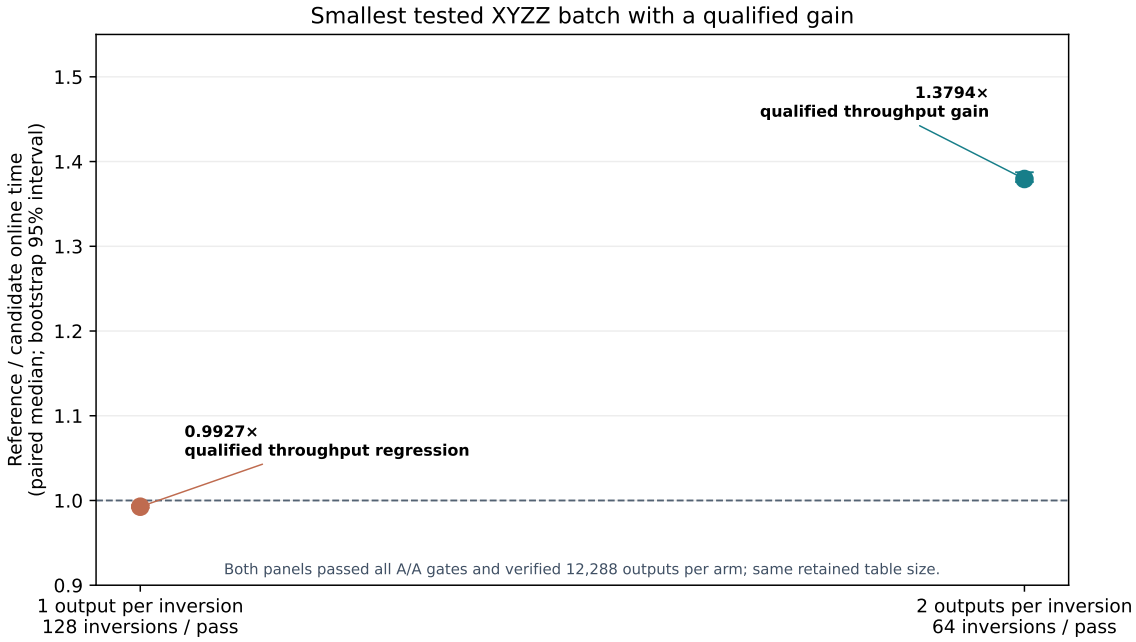
Rebuild with:

```
python3 experiments/prime-j0-batch-inversion-crossover-20261015/visuals.py
pandoc experiments/prime-j0-batch-inversion-crossover-20261015/
REPORT-2026-10-19.md \
  --from=gfm --pdf-engine=typst \
  --resource-path=experiments/prime-j0-batch-inversion-crossover-20261015 \
  -V geometry:margin=0.75in \
  -o experiments/prime-j0-batch-inversion-crossover-20261015/
REPORT-2026-10-19.pdf
```

Chunked inversion diagram (SVG) · PDF



Paired crossover graph (SVG) · PDF



Decision and next step

The evidence selects size 2 for the measured 128-output stream, with a qualified regression at size 1 and uncalibrated direct screen values for the larger sizes. The next step is a workload-matched measurement before applying this batch size outside the same fixture and Tier 1 instance envelope.